A twenty-two-variable candidate equation

Positive integer unknowns; degree 10; a 114-operation evaluator.

The only unknowns are the twenty-two distinct positive integers

$$J, F, \alpha, f, g, T, X, Y, \xi, \eta, \zeta, \upsilon, \gamma, \psi, \varphi, \epsilon,$$
$$\ell, n, Z, D, V, \delta \in \mathbb{Z}_{>0}.$$

Their ASCII names in the circuit are respectively `J, F, alpha, f, g, T, X, Y, xi, eta, zeta, upsilon, gamma, psi, varphi, epsilon, ell, n, Z, D, V, delta`. Each name denotes one unknown.

The four fixed integers are `beta`, `K`, `mu`, and `omega`, defined below. The single equation is

$$\boxed{\texttt{hDJ} = 0.}$$

Here `hDJ` is the last gate of the complete circuit on the next page. Every `h`-name denotes an integer polynomial abbreviation, not an additional unknown. All arithmetic in the circuit is over the integers.

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{47}, \quad B = \rho^{2011}, \quad \beta = B - 1,$$
$$G_m(x) = \sum_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{1005}.$$

$$\begin{aligned}\mu &= \beta - G_{951}(\rho)\\ &\quad - \rho^{960}G_{26}(\rho) - \rho^{1004},\\ \omega &= \beta + 8G_{917}(\rho) - 4 + (2^{45} - 4)\Lambda\\ &\quad + \rho^{960}G_9(\rho) + \rho^{987}G_8(\rho) + \rho^{1004},\\ K &= \rho^{18} + \rho^{1004} + \rho^{2020}\\ &\quad + 2\Lambda\Phi(1/\rho),\end{aligned}$$

where

$$\begin{aligned}\chi_i &= 1 + 2^{10}(1 + w_i + 2^8 n_i\\ &\quad + 2^{17}p_i + 2^{25}s_i) \quad (0 \le i \le 941),\end{aligned}$$
$$\pi(0) = 0, \quad \pi(i) = 1 + \sum_{j=1}^{941} \left\lfloor \frac{2^{44} - 1 + \chi_j - \chi_i}{2^{44}} \right\rfloor \quad (1 \le i \le 941).$$

$$\begin{aligned}\Phi(x) = &\sum_{i=0}^{941} \chi_i x^{\pi(i)}\\ &+ 2^{10}x^{969}(x^9 + 2^{17})G_8(2x)\\ &+ 2^{18}x^{942}(x^{18} + 2^{17})G_9(2x).\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 942 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. For $d = 0, 1$, let

$$h(q,b) = 2q + b + 2, \quad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\},$$
$$r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \dots, q-1\}|.$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 941$; all index ranges are fixed:

index i	tile $t_i = (w_i, n_i, p_i, s_i)$
0	$(2, 435, 3, 1)$
$j+1$ $(j = 2q + b < 430)$	$((1-d_j)r_{0,u_j}, h(q,b), d_j r_{1,u_j}, v_j)$
$433 - 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 432a + b)$
$\substack{423+2r_{d,q}+b \\ q\in\mathcal{S}_d,\ d,b=0,1}$	$(dr_{d,q}, b, (1-d)r_{d,q}, h(q,b))$
$923 + 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 438)$
927	$(2, 434, 2, 0)$
928	$(3, 436, 4, 4)$
929	$(4, 436, 4, 0)$
$496 + t$ $(t = 434, 435, 436)$	$(1, 438, 1, t)$
$933 + 3a + b$ $(a, b = 0, 1, 2)$	$(a^2, 437, b, 437)$

A closed decoder for the 430 actions. Put $D_b(A,k) = \lfloor A/b^k \rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$\mathcal{N} = \mathcal{A} \bmod 2^{636}, \quad \mathcal{T} = \lfloor \mathcal{A}/2^{636} \rfloor,$$
$$\nu_j = D_2(\mathcal{N}, j), \quad \kappa_j = (\sigma_j + 2)^{1-\nu_j} \quad (0 \le j < 430),$$
$$\sigma_j = \sum_{k=0}^{j-1} \nu_k, \quad \Pi_j = \prod_{k=0}^{j-1} \kappa_k \quad (0 \le j \le 430),$$
$$m_q = D_2(\mathcal{N}, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14 + m_q(D_{16}(\lfloor \mathcal{T}/\Pi_{430} \rfloor, \tau_q) - 14),$$
$$u_j = (\sigma_{j+1} - (\lfloor \mathcal{T}/\Pi_j \rfloor \bmod \kappa_j)) \bmod 216,$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums/products are 0/1. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$ in base 10. Its complete decimal expansion is the concatenation of these lines:

```
12889260410359561746736540743515845134244260640012702033212621
95398074180784058571518488282119861023093355389406939414042892
97455773992193784795705001618777025966369137954859764532832722
75411843410809038713502633204617333246733616940964578272360901
56186216508504953649192752154519268024509952082950238865468307
22999632482769253980274091217460707829023970817244790340704449
09085581271192071656170732004937337082171141082486384269072884
72778087419663417942892520175630649792329265834490253035577533
96449689808375788457488971017043689875433162972390177936123032
89025657757489526342497438772045714099661532481123050235898973
27443609903800541199351405985754278307227777584454508422717998
33411356440933534607489268032843 7
```

The fixed integer $\mathcal{A}$ above is written in base 10. This is a research candidate. Exact polynomial calculations certify degree 10 and the substitution identities. A Lean witness-transport theorem and an internal intuitionistic PRA equivalence for this candidate have not been proved. No minimality claim is made.

Complete 114-operation circuit

Each line defines one gate. The only inputs are the 22 positive integers and the four fixed integers on the first page. The final gate `hDJ` is a sum of 13 squares; setting it to zero is the displayed equation. Gate names are local abbreviations and are distinct from all input names.

```
1.  hA = J * beta
2.  hB = 1 + hA
3.  hC = hB * hB
4.  hD = hB * hC
5.  hE = Y * hC
6.  hF = hB - F
7.  hG = hF - alpha
8.  hH = X * Y
9.  hI = eta + zeta
10.  hJ = Y * hI
11.  hK = eta + hJ
12.  hL = hK * hK
13.  hM = hH + Y
14.  hN = 4 * hM
15.  hO = 3 + hN
16.  hP = hM * hM
17.  hQ = hO + hP
18.  hR = hM * hK
19.  hS = X + hR
20.  hT = hH * hJ
21.  hU = hD * hA
22.  hV = K * hB
23.  hW = X + hV
24.  hX = hG * hW
25.  hY = F * hB
26.  hZ = hX - hY
27.  hAA = ell + hI
28.  hAB = hAA + xi
29.  hAC = hI * hZ
30.  hAD = hAB - hAC
31.  hAE = psi * hAD
32.  hAF = Y * hA
33.  hAG = hZ * hD
34.  hAH = hAE + hAF
35.  hAI = hAH - hAG
36.  hAJ = Z * Z
37.  hAK = psi * psi
38.  hAL = ell * hAA
39.  hAM = 1 + hAL
40.  hAN = hAK * hAM
41.  hAO = hAJ - hAN
42.  hAP = hF * hA
43.  hAQ = alpha + hAP
44.  hAR = hC - 1
45.  hAS = hAQ * hAR
46.  hAT = omega * hB
47.  hAU = mu + hAT
48.  hAV = J * hAU
49.  hAW = hAS + hAV
50.  hAX = hI - epsilon
51.  hAY = hAX - 1
52.  hAZ = varphi * gamma
53.  hBA = hH * hB
54.  hBB = hAZ + hBA
55.  hBC = hBB - hAY
56.  hBD = varphi * hS
57.  hBE = hO * hBC
58.  hBF = hBD + hBE
59.  hBG = D * D
60.  hBH = varphi * varphi
61.  hBI = delta * n
62.  hBJ = 1 + hBI
63.  hBK = hBH * hBJ
64.  hBL = hBG - hBK
65.  hBM = f * f
66.  hBN = hBM - 1
67.  hBO = delta * hBN
68.  hBP = n * g
69.  hBQ = hBP * hBP
70.  hBR = hBQ - hBO
71.  hBS = T * hK
72.  hBT = hBS - epsilon
73.  hBU = hBM * hBT
74.  hBV = hBU - hK
75.  hBW = upsilon * upsilon
76.  hBX = V * V
77.  hBY = hBW - hBX
78.  hBZ = hBO * hBY
79.  hCA = hBW - hBZ
80.  hCB = hCA - 1
81.  hCC = hAO * hAO
82.  hCD = hBL * hBL
83.  hCE = hCB * hCB
84.  hCF = hBR * hBR
85.  hCG = hCC + hCD
86.  hCH = hCG + hCE
87.  hCI = hCH + hCF
88.  hCJ = psi - hU
89.  hCK = hCJ * hCJ
90.  hCL = hCI + hCK
91.  hCM = varphi - hE
92.  hCN = hCM * hCM
93.  hCO = hCL + hCN
94.  hCP = epsilon - hAW
95.  hCQ = hCP * hCP
96.  hCR = hCO + hCQ
97.  hCS = ell - hT
98.  hCT = hCS * hCS
99.  hCU = hCR + hCT
100.  hCV = n - hL
101.  hCW = hCV * hCV
102.  hCX = hCU + hCW
103.  hCY = Z - hAI
104.  hCZ = hCY * hCY
105.  hDA = D - hBF
106.  hDB = hDA * hDA
107.  hDC = V - hBV
108.  hDD = hDC * hDC
109.  hDE = delta - hQ
110.  hDF = hDE * hDE
111.  hDG = hCX + hCZ
112.  hDH = hDG + hDB
113.  hDI = hDH + hDD
114.  hDJ = hDI + hDF
```